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Admittance spectroscopy for organic materials with Gaussian density of states

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ABSTRACT

This work presents a systematic investigation of admittance spectra for single-carrier devices based on organic semiconductors featuring a Gaussian density of states. The dc current density is first numerically calculated using the master-equation-based mobility model. The small-signal admittance is subsequently computed from the first-order small-signal transport equations under the corresponding dc bias. The peak frequency f_{max} of the negative differential susceptance is systematically compared with the dc transit time τ. It is demonstrated that within the typical bias range, the product $f_{max}\tau$ exhibits a weak dependence on both the applied bias voltage and the energetic disorder width, and remains nearly constant at approximately 0.69, close to the classical Mott–Gurney result for a device with constant carrier mobility. The combined analysis of admittance spectra and steady-state log–log current–voltage characteristics provides an effective approach to narrow the possible range of mobility parameters in disordered organic semiconductors.

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I. INTRODUCTION

Charge transport in organic semiconductors is usually characterized by the carrier mobility, which is important for organic optoelectronic devices such as light-emitting diodes, field-effect transistors, and photovoltaic cells.[1–4] In single-carrier devices with Ohmic injection, the current density–voltage characteristics are often interpreted using space-charge-limited current (SCLC) theory.[5–7] For a constant mobility, the Mott–Gurney law gives a quadratic voltage dependence. Admittance spectroscopy also provides an independent frequency-domain measurement, in which the peak in $-\Delta B$ or in $-\mathrm{Im}\ Z$ has been successfully used to estimate the carrier transit time and mobility.[8–12]

In many organic semiconductors, energetic disorder has a strong influence on charge transport.[13–15] The motion of carriers occurs by hopping between localized states, and the mobility may depend on carrier concentration, electric field, temperature, and the energetic distribution of available states.[16–20] For an exponential density of states (EDOS), the mobility follows a power-law dependence on carrier concentration, and the relation between the admittance peak frequency and the transit time depends on the width of the distribution.[18,21,22] For a Gaussian density of states (GDOS), the mobility depends on carrier concentration and electric field in a more coupled manner.[16,17,23–25] The corresponding relation between the admittance characteristic frequency and the carrier transit time in the GDOS case has received less systematic study, which motivates the numerical analysis below.

A related difficulty appears in the numerical analysis of steady-state log–log J–V curves for the case of GDOS. The mobility is determined by the coupled effects of the intrinsic rate v_0, the Gaussian disorder width σ, and the lattice constant parameter a. These parameters jointly control the magnitude of the mobility and its dependence on carrier concentration and electric field and, therefore, determine the calculated log–log J–V characteristics. As a result, a change in one parameter may be compensated by changes in the others, so different parameter sets can give similar log–log J–V characteristics within a limited voltage range. This non-uniqueness arises because, under space-charge-limited conditions, both the carrier concentration and the electric field vary across the device and enter the calculated mobility.[16] As a result, steady-state log–log J–V fitting alone may not be sufficient to uniquely constrain the three parameters v_0, σ, and a.

In this work, numerical admittance calculations are carried out for single-carrier SCLC devices described by the mobility parameterization for a GDOS. The dc current density and electric field are obtained self-consistently from the drift-current equation and Poisson equation, and the small-signal admittance is

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calculated by solving the time-dependent partial differential electric-field equation. We first show that the same experimental log–log J–V curve can be reproduced using two different parameter sets. The peak frequency of $-\Delta B$ is then compared with the dc transit time, and the product $f_{max}\tau$ is evaluated for different bias voltages and disorder widths. In Sec. III, the mobility extracted from admittance spectra is used to examine whether the coupled GDOS parameters can be further constrained.

II. THEORY AND METHODS

A. GDOS mobility model

The present work considers localized-state hopping in disordered organic semiconductors. Delocalized band transport is not considered here. For an EDOS, an analytical mobility expression can be derived from the percolation model,[18] and the corresponding single-carrier admittance has also been analyzed.[21,22] For a GDOS, numerical and Monte Carlo simulations have been used to describe the temperature, carrier-concentration, and electric-field dependences of carrier mobility.[13,16] Two widely used descriptions are the transport-energy model and the master-equation model.[16,26] The master-equation model gives explicit relationships among the three coupled mobility parameters. Therefore, we use this model to examine whether admittance measurements can provide additional constraints on these parameters.[16] According to the master-equation model, the mobility is written as the product of a carrier-concentration-dependent zero-field term and an electric-field-enhancement factor,

$$\mu(T,\ p,\ E) = \mu(T,\ p) f(T,\ E), \tag{1}$$

where T is the absolute temperature, p the local carrier concentration, E the local electric field, and $f(T,\ E)$ the electric-field-enhancement factor. The density-dependent term takes the form,

$$\mu(T,\ p) = \mu_0(T) \exp\left[\frac{1}{2}(\hat{\sigma}^2 - \hat{\sigma})(2pa^3)^{\delta}\right], \tag{2}$$

where $\sigma\hat{} = \sigma/(k_B T)$, with σ being the Gaussian disorder width and k_B the Boltzmann constant. The parameter a represents the spacing between neighboring sites in the idealized regular cubic lattice used in the master-equation model.[16] For polymeric materials, this fitted lattice constant should be regarded as an effective parameter rather than a specific microscopic structural distance, because the model parameters effectively represent an average over intra- and interchain hopping processes.[16] The temperature-dependent prefactor $\mu_0(T)$ and the mobility-scale prefactor μ_0 are defined as[16]

$$\mu_0(T) = \mu_0 c_1 \exp\left(-c_2 \hat{\sigma}^2\right), \quad \mu_0 = \frac{a^2 \nu_0 q}{\sigma}, \tag{3}$$

where q is the elementary charge. The mobility prefactor μ_0 is expressed in $m^2\,V^{-1}\,s^{-1}$. The constants c_1 and c_2 in the GDOS parameterization are 1.8×10^{-9} and 0.42, respectively. The exponent δ is given by

$$\delta = 2\frac{\ln(\hat{\sigma}^2 - \hat{\sigma}) - \ln(\ln 4)}{\hat{\sigma}^2}. \tag{4}$$

The electric-field-dependent factor can be written as

$$f(T, E) = \exp\left\{0.44(\hat{\sigma}^{3/2} - 2.2)\left[\sqrt{1 + 0.8\left(\frac{qaE}{\sigma}\right)^2} - 1\right]\right\}. \tag{5}$$

Since ν_0 enters the mobility through μ_0, the analysis below treats μ_0, σ, and a as the three coupled mobility parameters. Their physical roles are separated explicitly: μ_0 controls the overall mobility magnitude, σ determines the variation of mobility with carrier concentration, and a mainly affects the high-field mobility enhancement. The mobility parameterization is first applied to calculate the steady-state current-density characteristics. Based on the calculated spatial distributions of carrier concentration $p_{dc}(x)$ and electric field $E_{dc}(x)$, where x is the distance from the injecting electrode, the GDOS admittance properties are then evaluated.

B. Steady-state log–log J–V calculation

A unipolar hole-only device of thickness L is considered under a dc bias V_{dc}. At low bias, drift and diffusion can both contribute to the current.[27,28] Admittance measurements for determining carrier mobility are seldom carried out in this range because it is difficult to distinguish the mechanisms governing carrier transport. Therefore, the quantitative admittance analysis in this work focuses on biases above approximately 3 V, corresponding to regions II and III defined below, where drift transport dominates.[5–7,27,28] The steady-state current density is

$$J_{dc} = q\, p_{dc}(x)\, \mu[T,\ p_{dc}(x),\ E_{dc}(x)]\, E_{dc}(x), \tag{6}$$

and the electrostatic field obeys Poisson's equation,

$$\frac{dE_{dc}(x)}{dx} = \frac{q\, p_{dc}(x)}{\varepsilon}, \tag{7}$$

where ε is the permittivity of the organic layer. For a prescribed J_{dc}, Eqs. (6) and (7) are solved numerically. The applied voltage is then obtained from

$$V_{dc} = \int_0^L E_{dc}(x)dx. \tag{8}$$

To calculate the dc profiles, the layer is divided into 10 000 uniformly spaced points. For $L = 560$ nm, this gives $\Delta x \approx 0.056$ nm. Starting from a small value of $E(0)$ at the injecting electrode, $E(x)$ is calculated point by point for a given J_{dc}. At each spatial point, $p(x)$ is obtained from the finite-difference form of Eq. (7), and the next electric-field value is determined by bisection so that Eq. (6) is satisfied using the local GDOS mobility. The calculated field is then integrated according to Eq. (8) to obtain V_{dc}. Thus, for a given J_{dc} and fixed mobility parameters, the calculation

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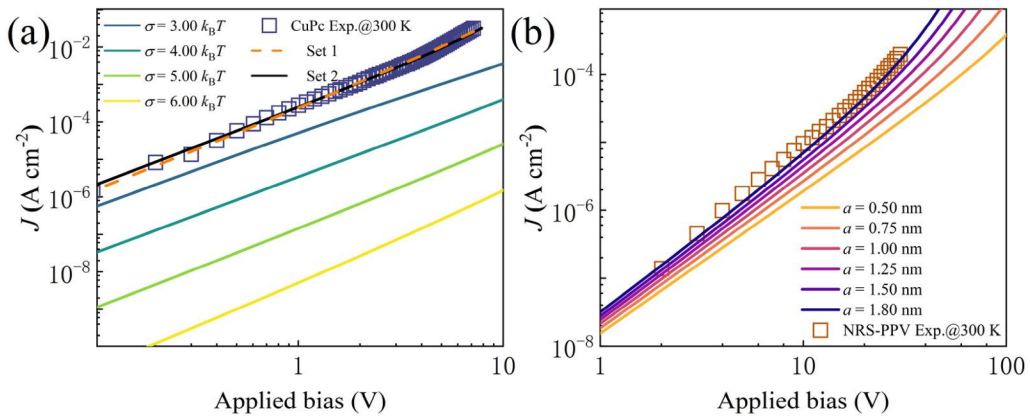


FIG. 1. Steady-state log–log J–V characteristics. (a) Calculated J–V curves for $\sigma/(k_BT) = 3$, 4, 5, and 6, together with the CuPc experimental data.[29] The black solid and orange dashed curves are calculated using parameter sets 1 and 2, respectively. (b) Calculated J–V curves for different a, together with the NRS-PPV experimental data.[16]

directly gives the corresponding V_{dc} and dc spatial profiles. When an experimental J–V characteristic is fitted, the measured data are fixed and the mobility parameters μ_0, σ, and a are adjusted.

The calculated log–log J–V curves in Fig. 1(a) first illustrate the effect of energetic disorder by varying σ/k_BT from 3 to 6, with fixed values of $\mu_0 = 72\ m^2\ V^{-1}\ s^{-1}$ and $a = 1.80$ nm. As shown in Fig. 1(a), with increasing σ, the slope of the log–log J–V curves increases slightly, whereas the magnitude of the current density decreases significantly.

Figures 1(a) and 1(b) also show the room-temperature experimental log–log J–V data for copper phthalocyanine (CuPc) and poly[4′-(3,7-dimethyloctyloxy)-1,1′-biphenylene-2,5-vinylene] (NRS-PPV), respectively. The CuPc and NRS-PPV experimental data are taken from Refs. 29 and 16, respectively. Specifically, the CuPc data were measured in an indium tin oxide (ITO)/CuPc (200 nm)/Au hole-only device,[29] whereas the NRS-PPV data were measured in an ITO/NRS-PPV (560 nm)/Au hole-only device.[16] By contrast, the curves in Fig. 1 are calculated in this work using Eqs. (6)–(8). In Fig. 1(b), $\mu_0 = 72\ m^2\ V^{-1}\ s^{-1}$, $\sigma = 0.14$ eV, and $L = 560$ nm are taken from Ref. 16, while a is varied. For the discussion below, the bias range is divided into three regimes, denoted regions I, II, and III, according to the slopes of the experimental log–log J–V characteristics. Region I corresponds to bias voltages below approximately 3 V, where drift and diffusion can both contribute to the current. In region II, the current density exhibits an approximately constant slope, indicating that the mobility is mainly affected by carrier concentration, while the influence of the electric field remains relatively weak. In region III, at higher bias voltages, the slope of the log–log J–V curve increases with increasing voltage, suggesting that the electric-field dependence of the mobility can no longer be neglected. The log–log J–V curve of CuPc at 300 K is fitted with two different parameter sets: parameter set 1 ($\mu_0 = 17\,500\ m^2\ V^{-1}\ s^{-1}$, $\sigma = 0.140$ eV, $a = 1.30$ nm) and parameter set 2 ($\mu_0 = 750\ m^2\ V^{-1}\ s^{-1}$, $\sigma = 0.115$ eV, $a = 2.00$ nm).[16,29] Despite the substantial difference in parameter values, both parameter sets give reasonably good fits to the measured log–log J–V characteristics. These two parameter sets illustrate the degree of non-uniqueness arising from the coupling among μ_0, σ, and a in steady-state log–log J–V fitting. Figure 1(b) further shows that varying a primarily modifies the high-bias bending region. The onset of this slope enhancement shifts to lower bias voltages as a increases. These comparisons demonstrate that steady-state log–log J–V fitting alone cannot uniquely determine μ_0, σ, and a; mobility extracted from admittance spectra can, therefore, provide an additional constraint.

C. Spatial profiles and transit time calculation

According to Poisson's equation, the electric field, carrier concentration, and mobility are intrinsically coupled, so their spatial distributions can vary under different bias conditions. Figure 2 presents the calculated spatial profiles of these quantities at two representative bias voltages. The carrier concentration and mobility in Fig. 2 are given in SI units, m^{-3} and $m^2\ V^{-1}\ s^{-1}$, respectively. The parameters are the same as those used in the previous fitting of the experimental data in Fig. 1(b), and the bias voltages of 7 and 20 V are chosen as representative values in regions II and III, respectively. The higher bias leads to a larger carrier concentration and a stronger electric field throughout the device. Because the mobility depends on both $p_{dc}(x)$ and $E_{dc}(x)$, the mobility profile is spatially non-uniform even under a constant dc current density. Interestingly, in contrast to the case at 7 V, the spatial distribution of mobility at 20 V first decreases and then increases. The dashed curves in Figs. 2(a)–2(c) show the 20 V results obtained with $f_{TE} = 1$, corresponding to the case where the electric-field enhancement of the mobility is removed. Under this condition, the $p(x)$ and $E(x)$ profiles change only slightly, whereas the mobility comparison indicates that the high-bias increase in $\mu(x)$ originates from its electric-field dependence.

The inter-electrode transit time is given by

$$\tau = \int_0^L \frac{dx}{\mu[T, p_{dc}(x), E_{dc}(x)]E_{dc}(x)}. \quad (9)$$

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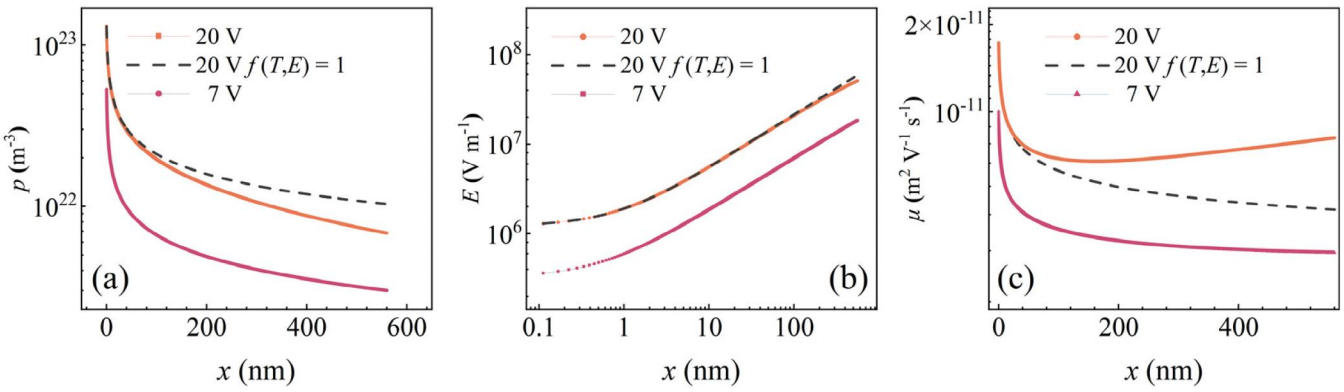


FIG. 2. Calculated spatial profiles at $V_{dc} = 7$ and 20 V: (a) hole concentration $p_{dc}(x)$, (b) electric field $E_{dc}(x)$, and (c) mobility $\mu(x)$. The dashed curves show the 20 V results calculated with $f_{TE} = 1$. Other parameters are taken from Ref. 16 and are the same as those used in Fig. 1(b).

In the following discussion, the transit time will be correlated with the characteristic frequency of the admittance response and used to extract the carrier mobility. Therefore, the mobility extracted from admittance spectra, μ_{AS}, can be expressed as

$$\mu_{AS} = \frac{L^2 f_{max}}{\eta V_{dc}}, \tag{10}$$

where η is defined by $f_{max}\tau = \eta$. Since the mobility varies spatially across the device, the spatially averaged mobility is defined as

$$\bar{\mu} = L^{-1}\int_0^L \mu(x)\mathrm{d}x. \tag{11}$$

Equation (11) gives the simple spatial average of the local mobility $\mu(x)$. This representative value is used when the spatially varying theoretical mobility is compared with the single admittance-derived mobility μ_{AS} obtained independently from Eq. (10). In the subsequent concentration- and field-dependence analyses, $p(x)$, $E(x)$, and $f_{TE}(x)$ are represented by their simple spatial averages $\bar{p}$, $\bar{E}$, and $\bar{f}_{TE}$, respectively; for the electric field, $\bar{E} = V_{dc}/L$.

To examine the influence of the averaging definition, we also evaluate the log average of a spatially varying quantity $X(x)$,

$$\bar{X}_{log} = \exp\left[\frac{1}{L}\int_0^L \ln X(x)\,\mathrm{d}x\right], \tag{12}$$

where X denotes any of p, μ, E, and f_{TE}. The log average is used as a sensitivity check of the representative spatial quantities. The numerical values depend on the averaging definition, while the simple spatial average is retained for the subsequent analysis to provide consistent representative values for the spatially varying quantities.

D. Frequency-domain admittance calculation

When a small sinusoidal voltage with a frequency $f = \omega/2\pi$ is superimposed on the dc bias, the time- and position-dependent electric field and carrier concentration, together with the time-dependent current density, are written as sums of their dc and small-signal ac components,

$$E(x, t) = E_{dc}(x) + e(x, t), \tag{13a}$$

$$p(x, t) = p_{dc}(x) + p_{ac}(x, t), \tag{13b}$$

$$J(t) = J_{dc} + j_{ac}(t). \tag{13c}$$

Here $e(x, t)$, $p_{ac}(x, t)$, and $j_{ac}(t)$ are the small-signal perturbations of the electric field, carrier concentration, and current density, respectively. The mobility $\mu(x, t)$ is evaluated from Eqs. (1)–(5) using $p(x, t)$ and $E(x, t)$. The total current density is written as[8,9]

$$J(t) = q\,p(x,\ t)\,\mu(x,\ t)\,E(x,\ t) + \varepsilon\frac{\partial E(x,\ t)}{\partial t}. \tag{14}$$

Substituting Eq. (13) into Eq. (14), retaining only the first-order terms, and applying a Fourier transform gives the small-signal admittance response under the corresponding dc operating condition. In the frequency domain, $j_{ac}(t)$ and $e(x, t)$ are transformed into $j_{ac}(\omega)$ and $e(x, \omega)$, respectively. The ac voltage is obtained from

$$v_{ac}(\omega) = \int_0^L e(x, \omega)\mathrm{d}x. \tag{15}$$

The complex admittance per unit area is then written as

$$Y(\omega) = \frac{j_{ac}(\omega)}{v_{ac}(\omega)} = G(\omega) + \mathrm{i}B(\omega) = G(\omega) + \mathrm{i}\omega C(\omega), \tag{16}$$

where $G(\omega)$, $B(\omega)$, and $C(\omega)$ are the conductance, susceptance, and capacitance per unit area, respectively. Therefore, the capacitance is calculated as $C(\omega) = B(\omega)/\omega$, and the geometric capacitance per unit area is $C_{geo} = \varepsilon/L$. Figure 3 displays the normalized

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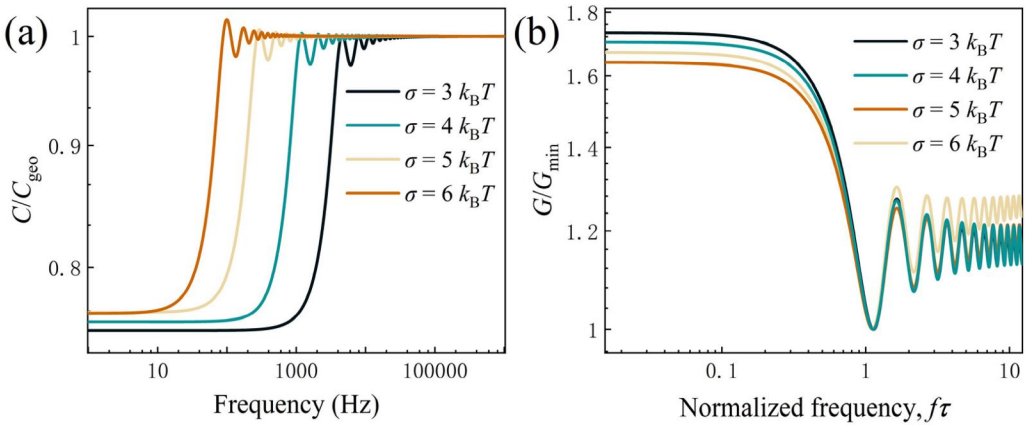


FIG. 3. Calculated admittance spectra at $J = 10^{-1}$ A m^{-2} for $\sigma/(k_BT) = 3$, 4, 5, and 6: (a) C/C_{geo} as a function of f and (b) G/G_{min} as a function of $f\tau$. The corresponding dc voltages are 0.43, 1.66, 6.59, and 19.46 V. Other parameters are taken from Ref. 16 and are the same as those used in Fig. 2. G_{min} is the minimum conductance of each spectrum.

capacitance C/C_{geo} and normalized conductance G/G_{min} calculated at a fixed current density of $J = 10^{-1}$ A m^{-2} for $\sigma/k_BT = 3$, 4, 5, and 6, corresponding to $V_{dc} = 0.43$, 1.66, 6.59, and 19.46 V, respectively.[11,30–33] The 0.43 and 1.66 V spectra are shown to illustrate the calculated frequency response of the present model at low bias; the quantitative mobility analysis below focuses on biases in regions II and III. As shown in Fig. 3, a negative contribution to the device capacitance appears at low frequencies. The characteristic frequency at which this negative contribution vanishes shifts to higher frequencies with increasing bias voltage. At high frequencies, the oscillations in Fig. 3(b) originate from the carrier transit time effect of the SCLC response. Kassing showed that this transit time effect produces periodic oscillations of the conductance and capacitance in SCLC diodes at high frequencies.[34] Ramachandhran *et al.* noted that, at high frequencies, the ac current density oscillates with a spatial period much smaller than the layer thickness, so diffusion is expected to damp these oscillations; they also pointed out that no such oscillations were observed experimentally in PPV-based hole-only devices.[21]

III. RESULTS AND DISCUSSION

A. Characteristic frequency and the approximate $f_{max}\tau$ coefficient

The $-\Delta B$ and $-\mathrm{Im}\ Z$ methods have been used to extract charge-carrier mobility.[8–11] For simplicity, the following analysis focuses mainly on $-\Delta B$, and the corresponding analysis based on $-\mathrm{Im}\ Z$ is analogous. The quantity $-\Delta B$ is defined relative to the geometric-capacitance baseline as[9,11]

$$-\Delta B = -\omega\left[C(\omega) - C_{geo}\right] = \omega\left[C_{geo} - C(\omega)\right]. \quad (17)$$

Figures 4(a) and 4(b) show the frequency-dependent $-\Delta B$ and $-\mathrm{Im}\ Z$, respectively, calculated at several dc bias voltages. To make our calculation more relevant for real organic materials, the bias voltages are selected from the fitting curve of the experimental data in Fig. 1(b), and all other parameters are kept identical to those used in the corresponding log–log J–V calculation. The frequency at which $-\Delta B$ reaches its maximum is related to the carrier transit time.[8,9] Since the transit time varies with dc bias, we examine the dimensionless product $f_{max}\tau$. The $-\Delta B$ peak is associated with carrier transit through the device, so f_{max} varies approximately with $1/\tau$. Previous studies of EDOS systems showed that $f_{max}\tau$ depends on the DOS width.[21,22] To examine the GDOS case, the capacitance spectra are calculated at a series of dc bias voltages spanning regions II and III, as shown in Fig. 4(c). For $\sigma = 0.140$ eV, the capacitance curves nearly intersect at $f_{max}\tau \approx 0.697$, with comparable results for $\sigma = 0.080$ and 0.110 eV ($f_{max}\tau \approx 0.705$ and 0.686, respectively). The inset shows the small deviations among these curves, indicating that the intersections are not exactly identical. This differs from the EDOS behavior reported in Refs. 21 and 22. In the present GDOS case, the mobility is mainly concentration dependent in region II, whereas in region III, it is affected by both carrier concentration and electric field. Figure 4(d) also shows the raw f_{max} values on the right axis for $\sigma = 0.080$, 0.110, and 0.140 eV. These values change with both applied bias and σ. Nevertheless, the $f_{max}\tau$ values extracted from $-\Delta B$ peaks remain nearly unchanged over a broad voltage range for the three σ values. Therefore, $f_{max}\tau \approx 0.69$ is used in the following analysis as a practical conversion coefficient for mobility extraction within the GDOS parameter range examined. This coefficient is close to the classical Mott–Gurney result of about 0.72, even though the present case involves concentration- and field-dependent mobility rather than a constant mobility.

B. Application to CuPc: Reducing the ambiguity of coupled GDOS parameters

As discussed above, in the GDOS mobility model, the three mobility-related parameters, namely, the lattice constant a, the

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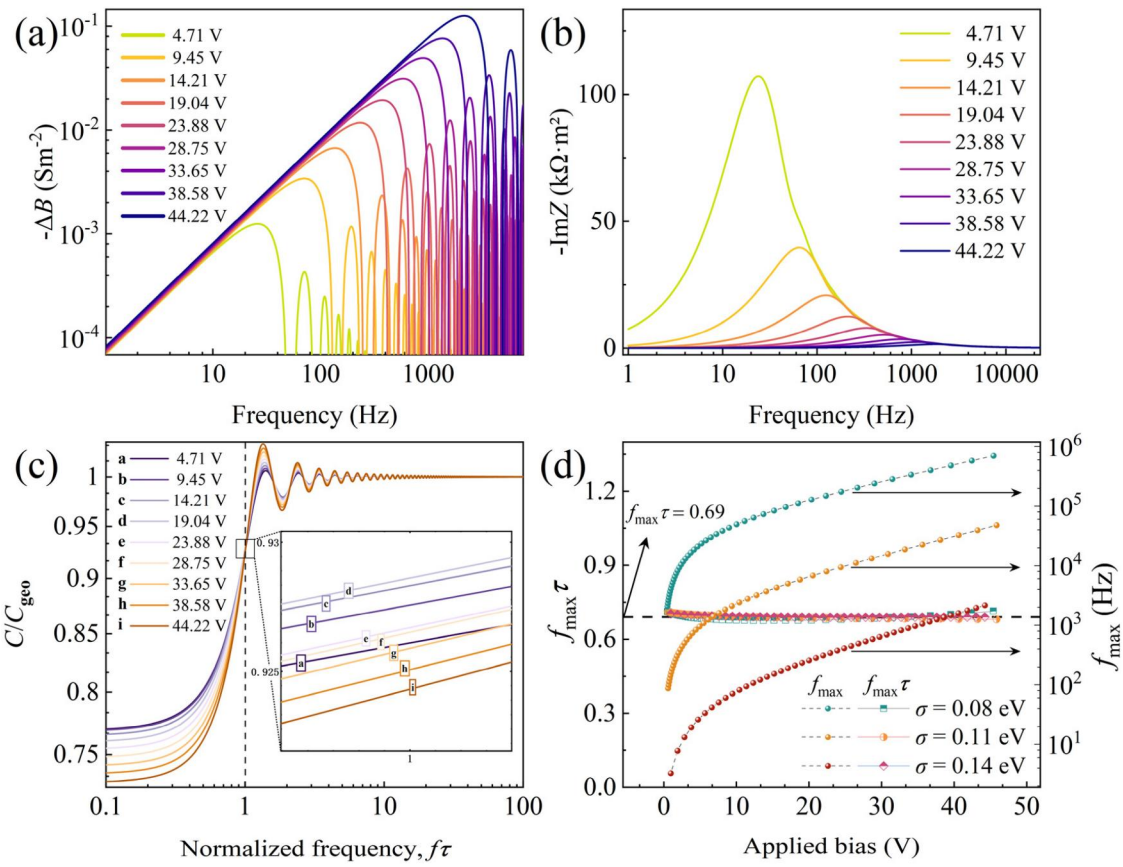


FIG. 4. (a) $-\Delta B$ and (b) $-\mathrm{Im}\,Z$ as functions of frequency at selected dc biases. (c) C/C_{geo} as a function of $f\tau$; the inset enlarges the intersection region. (d) $f_{max}\tau$ (left axis) and f_{max} (right axis) as functions of dc bias for $\sigma = 0.080$, 0.110, and 0.140 eV. The dashed line marks $f_{max}\tau = 0.69$; the arrows indicate the right-axis scale for f_{max}.

energetic disorder width σ, and the mobility prefactor μ_0, are strongly coupled. As shown in Fig. 1(a), two different parameter sets can provide comparably good fits to the CuPc experimental log–log J–V data, although the corresponding intrinsic rates ν_0 differ by a factor of about 67. To examine whether admittance spectroscopy, when combined with the room-temperature log–log J–V characteristics, can further constrain these parameters, the CuPc data in Fig. 1(a) are taken as a representative example. The same parameter sets used for the dc log–log J–V fitting are adopted in the admittance calculation, and the corresponding carrier mobility values are extracted from the $-\Delta B$ peaks. The J–V curve of CuPc is selected because its temperature-dependent characteristics have been successfully fitted in the literature using the master-equation mobility parameterization, which provides a reliable reference for comparison.[16,29]

Within region II, where the log–log J–V characteristic has an approximately constant slope, the mobility is mainly governed by carrier concentration, while the electric-field effect is relatively weak. Therefore, the admittance response is calculated at a series of dc biases within region II, and the mobility extracted from the peak frequency of $-\Delta B$ is plotted as a function of the corresponding carrier concentration in Fig. 5(a). For each dc bias, the corresponding dc solution is first used as the operating state for the small-signal admittance calculation. For each bias, f_{max} is determined from the maximum of $-\Delta B(f)$, and the admittance-derived mobility μ_{AS} is calculated from Eq. (10) using $\eta \approx 0.69$. For comparison, the mobility calculated directly from the GDOS mobility expression using the corresponding spatially averaged carrier concentration is also given. It can be seen that the mobility obtained from admittance analysis closely follows the carrier-concentration dependence described by Eq. (2). The increase in mobility with carrier concentration in Fig. 5(a) is consistent with Eq. (2), for which the concentration-dependent term increases with p for the present GDOS parameters.[16,26]

For region III, where the mobility is affected by both carrier concentration and electric field, admittance calculations are performed at a series of dc biases within this region. To examine whether the mobility extracted from admittance spectra can separately reproduce the field-dependent mobility trend, the concentration-dependent contribution corresponding to each bias

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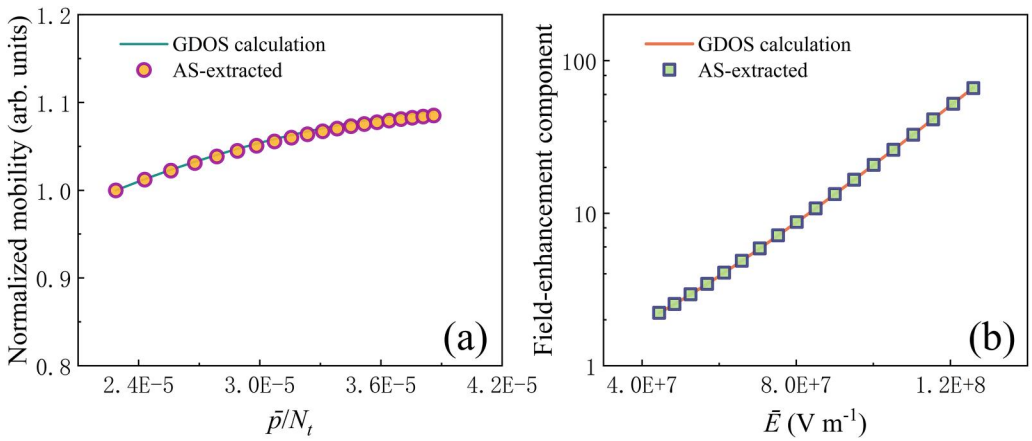


FIG. 5. Comparison between the GDOS mobility calculation and admittance analysis using parameter set 2. (a) Normalized concentration-dependent mobility as a function of $\bar{p}/N_t$ for dc biases in region II. (b) Field-dependent mobility component as a function of $\bar{E}$ for dc biases in region III. The AS-extracted values are obtained from the $-\Delta B$ peak using $f_{max}\tau \approx 0.69$.

voltage is divided out, and the remaining mobility component is plotted as a function of the device-averaged electric field $\bar{E}$ in Fig. 5(b). The local field-enhancement factor $f_{TE}(x)$ is calculated from Eq. (5) using the local electric field $E(x)$, and its simple spatial average is also shown for comparison. Likewise, the mobility extracted from admittance spectra closely reproduces the field-dependent mobility trend. These comparisons show that admittance spectroscopy can reflect the carrier-concentration and electric-field dependence of mobility in the GDOS case.

Figure 6(a) compares the carrier-concentration dependence of the mobility μ_{AS} extracted from admittance spectra using the parameter sets related to the J–V fitting in Fig. 1(a). The μ_{AS} values are calculated from the peak frequency of $-\Delta B$ using $f_{max}\tau \approx 0.69$. Notably, although parameter set 1 and parameter set 2 reproduce similar J–V characteristics, they lead to clearly different μ_{AS}–p trends. Figure 6(a) displays the pronounced influence of σ on the concentration dependence of mobility. For comparison, we also calculated the concentration dependence of mobility using the same σ but different values of a. Among the three parameter sets, the set with $\sigma = 0.115$ eV and $a = 1.30$ nm gives relatively better agreement with the experimental data. For the curve calculated using parameter set 2, $a = 2$ nm is near the upper limit of its possible range, yet the calculated mobility shows only a small difference, indicating that the concentration dependence of mobility is more sensitive to σ than to a. Thus, Fig. 6(a) is mainly useful for estimating σ and provides only limited information on a.

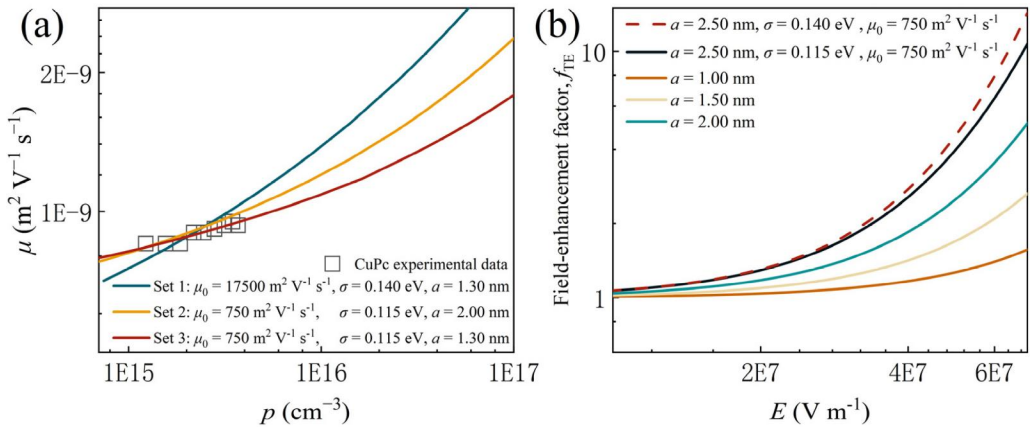


FIG. 6. Carrier-concentration-dependent mobility and field-enhancement analysis for the CuPc data. (a) μ_{AS} as a function of carrier concentration for parameter sets 1–3; symbols show the CuPc experimental data. The three calculated curves are scaled by factors of 2.36, 1.90, and 2.20, respectively. (b) f_{TE} as a function of E for different values of a and σ.

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To further examine the high-field behavior, Fig. 6(b) shows the field-enhancement factor f_{TE} as a function of electric field for different values of the lattice constant a. The stronger increase in f_{TE} at larger a indicates that the high-field mobility enhancement is sensitive to the lattice constant parameter. To compare the influence of σ and a over the electric-field range corresponding to the high-bias region III, we fixed a and μ_0 and calculated the f_{TE}–E curves for different σ values. The dashed curves show that varying the distribution width σ only slightly changes f_{TE}. Therefore, once σ has been estimated from the concentration-dependent mobility, mobility measurements in this high-bias range can be used to further estimate a. However, to the best of our knowledge, there are currently no available experimental data that simultaneously provide the high-bias mobility and the corresponding J–V characteristics. From the above analysis, a combined investigation of the p-dependent mobility together with the E-dependent mobility at high bias can help narrow the possible ranges of μ_0, σ, and a in the mobility expression.

IV. CONCLUSION

In this work, the admittance spectral characteristics of organic semiconductor devices described by the master-equation hopping model for a GDOS are investigated through numerical calculations. The raw peak frequency f_{max} changes with applied bias and σ, whereas the dimensionless product $f_{max}\tau$ remains nearly constant at approximately 0.69. On this basis, we further examine whether admittance spectroscopy can provide additional constraints on the mobility-related parameters. The results show that the admittance responses in different bias regions identified from the log–log J–V characteristics contain complementary information on concentration- and field-dependent transport. Therefore, combining admittance spectroscopy with dc log–log J–V fitting can help narrow the possible ranges of μ_0, σ, and a.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Guoliang Zhou: Formal analysis (lead); Investigation (lead); Software (lead); Visualization (lead); Writing – original draft (lead). **Shuang Li**: Formal analysis (supporting). **Lei Wang**: Formal analysis (supporting); Methodology (supporting); Writing – review & editing (supporting). **Tianyou Zhang**: Conceptualization (lead); Funding acquisition (lead); Methodology (supporting); Project administration (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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